Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 34)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 1 . 1 Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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0 1 . 2 Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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0 1 . 3 State the equation used to forecast a change in population size.

[1 mark]

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0 1 . 4 Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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0 2 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 2 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 2 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

Blue whale .....

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Large blue butterfly .....

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0 3 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

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0 3 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

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0 3 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

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0 3 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

0 3 . 5 Give **two** agricultural sources of methane.

[2 marks]

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0 4 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 4 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

0 4 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

0 4 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 4 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 4 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

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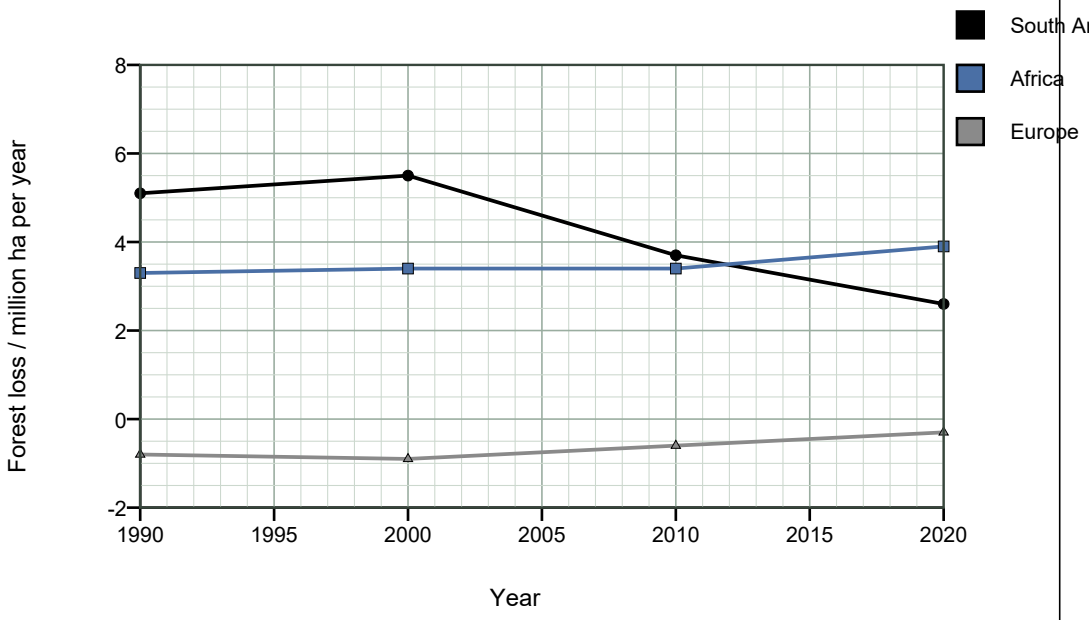
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0 5 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 5 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 5 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

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0 5 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

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0 5 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 6 . 1 Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 6 . 2 Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 6 . 3 Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 6 . 4 Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

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0 7 . 1 Give **two** wild species with potential as new food species and, for each, one characteristic that makes it useful.

[2 marks]

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0 7 . 2 Explain why indigenous species may be better for domestication than introduced livestock or crops.

[2 marks]

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0 7 . 3 Explain how species of the spinach family could help to solve a problem caused by irrigation.

[1 mark]

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0 8 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 8 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

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0 8 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

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0 8 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

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0 9 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 9 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size .....

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Distribution .....

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Survival rate .....

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Age structure .....

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0 9 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

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1 0 . 1 Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

1 0 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2 Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS